

Deep Learning based Alzheimer Disease Di- agnosis using Neuroimaging Data

Project report submitted in partial fulfillment
of the requirements for the degree of

Bachelor of Technology
in
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by

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CERTIFICATE

This is to certify that the project entitled “Deep Learning based Alzheimer Disease Diagnosis using Neuroimaging Data” , submitted by Aryan Dadhich(23ucs547) and Aviral Saini(23ucs551) in partial fulfillment of the requirement of degree in Bachelor of Technology (B. Tech), is a bonafide record of work carried out by them at the Department of Electronics and Communication Engineering, The LNM Institute of Information Technology, Jaipur, (Rajasthan) India, during the academic session 2023-2027 under my supervision and guidance and the same has not been submitted elsewhere for award of any other degree. In my/our opinion, this report is of standard required for the award of the degree of Bachelor of Technology (B. Tech).

Date

Adviser: Dr. Rahul Sharma

Abstract

Alzheimer’s Disease (AD) is a progressive neurodegenerative disorder characterized by structural brain deterioration, cognitive impairment, and functional decline. Early and accurate diagnosis is essential for timely intervention and improved clinical outcomes. In this project, we present a deep learning–based pipeline for multi-class classification of Alzheimer’s disease stages—Cognitively Normal (CN), Early Mild Cognitive Impairment (EMCI), Mild Cognitive Impairment (MCI), and Alzheimer’s Disease (AD)—using structural MRI data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI).

The proposed methodology includes a comprehensive neuroimaging preprocessing workflow that performs denoising, skull stripping, N4 bias-field correction, spatial registration to the MNI-152 anatomical template, intensity normalization, and multi-view slice extraction across axial, sagittal, and coronal planes. After rigorous preprocessing, several deep learning architectures were evaluated, including custom CNN models (CNN-5, CNN-10, CNN-15) and state-of-the-art ImageNet-pretrained models (EfficientNet-B3, ResNet50, and ResNet101). These models were trained and tested on a curated dataset of 18,030 MRI slices distributed evenly across four diagnostic classes.

Experimental results demonstrate a clear performance gap between shallow CNNs and deeper transfer-learning architectures. While the custom CNN models achieved accuracies between 31% and 59%, pretrained models substantially outperformed them. EfficientNet-B3 achieved 52% accuracy and a validation AUC of 0.80, whereas ResNet50 reached 50% accuracy with notable improvements in class separability. Among all models, **ResNet101 exhibited the highest performance**, achieving an accuracy of **91%**, macro F1-score of **0.91**, and per-class AUC values ranging from **0.98 to 0.99**. The confusion matrix further confirmed strong class-wise discrimination with minimal misclassification.

The results highlight that deeper residual networks, particularly ResNet101, are highly effective for MRI-based AD stage classification due to their superior feature extraction capability and stable gradient propagation. This work demonstrates the potential of deep learning to enhance neurodegenerative disease diagnosis and serves as a strong foundation for future developments. In the next phase, we aim to extend this system into a *multi-modal diagnostic framework* by incorporating genetic biomarkers, cognitive scores, demographic variables,

and additional clinical data to build a more holistic and robust Alzheimer's disease prediction model.

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Chapter 1

Introduction

1.1 The Area of Work

Magnetic Resonance Imaging(MRI) is an advanced technique used in medical to produce high resolution images of organs and soft tissues of our body.It works by aligning hydrogen protons with the strong magnetic field, when proton return to there initial state they emit signals which are then captured and converted into cross-Sectional and 3D images.

In Research and diagnosis purposing MRI images are used widely since they visualize structural brain changes without ionizing radiations. It plays a crucial role in diagnosing neurological disorders, including Alzheimer's disease, brain tumors, multiple sclerosis, and stroke.

1.2 Problem Addressed

Alzheimer Disease is a neurological disorder which causes gradual decline in memory , thinking ability and daily functioning .This is due to accumulation of certain proteins in brain such as amyloid- plaques and tau tangles which damage neurons breaking communication between brain cells , structural changes which observed are cortical thinning, enlargement of brain ventricles and hippocampus shrinkage.Early identification through MRI helps in diagnosis, monitoring disease progression, and improving research on automated classification using deep learning models.

A person can be classified into three major stages of Alzheimer CN(Congnitively Normal),MCI(Mild Cognitive Impairment) and AD(Alzheimer Disease).Our

main goal is to classify the patient situation or stage for this disease.

Brain Region	CN	MCI	AD
Hippocampus	Normal	starts to shrink	Severly atrophied
Entorihnal cortex	Normal	Early Degenration	Heavy loss
Parietal Lobes	Normal	Mild changes	Atrohpy
Frontal Lobes	Normal	Normal	Shrinkage
Posterior Cingulate	Normal	Reduces Matabolism	Significantly affected
Ventricals	Normal	Slightly enlarged	Markedly enlarged

Chapter 2

Literature Review

2.1 Introduction

Since the 1940s, extensive research has been conducted on Alzheimer’s Disease (AD) diagnosis, with neuroimaging-based studies gaining prominence in the past decade. The availability of high computational power has significantly enhanced automated diagnostic capabilities using Multimodal data. Alizadeh et al. [1] proposed a 1D-CNN model utilizing a functional atlas for rs-fMRI parcellation into 39 brain regions, achieving reasonable accuracy but limited generalization due to small dataset size. A deep ensemble approach integrating MRI and fMRI was introduced in [2], trained on four public datasets; however, class imbalance and model selection issues affected robustness. A multimodal framework was explored in [3] using three feature-selection techniques with conventional SVM, but lacked comparison with advanced SVM variants. Another study [4] employed ROI-based fusion of sMRI, diffusion MRI, and rs-fMRI, achieving decent performance yet constrained by limited data and traditional classifiers. Zheng et al. [5] reported disrupted functional connectivity in the default mode network and altered cerebral blood flow in AD patients (sensitivity 65.7% contrast, Wang et al. proposed a 3D-DenseNet ensemble to mitigate overfitting and enhance AD/MCI diagnosis using sMRI, while Li et al. [6] integrated 3D-CNN with LSTM to capture spatial-temporal features in neuroimages, though optimal LSTM configuration remains challenging. Overall, multimodal fusion of MRI and other Modalities provides complementary information, and ensemble deep learning frameworks exhibit higher robustness for AD classification. Furthermore, predicting converter vs. non-converter MCI states represents a promising direction for improved AD prognosis.

Chapter 3

Proposed Work

3.1 Dataset Description

The Alzheimer's Disease Neuroimaging Initiative (ADNI) is a organization which allows researchers to use there collected high quality MRI , PET , genitic and clinical data for study of particular disorders.

The dataset consists of 18,030 MRI slices categorized into four classes: AD, CN, EMCI, and MCI.This dataset is gathered by using middle 15 slices of each patient , before using for training it has been properly preprocessed to remove any mismatches such as registration.

3.2 Methodology

3.2.1 Data preprocessing

The ADNI dataset was initially present in DICOM (Digital Imaging and Communications in Medicine) format. Using SimpleITK they were converted to SimpleITK images which underwent prechecks such as skipping invalid/empty images as it prevents errors in later steps. All valid scans were first converted to standard formats by casting them to sitkFloat32 datatype and "IAL" (Inferior-Anterior-Left) orientation. These are needed for numerical stability in later operations. The scans were first denoised,

A head mask was formed for each of them by rescaling the intensities and setting threshold values. Fig(a)

The N4BiasFieldCorrectionImageFilter was applied to remove intensity non-uniformities after first shrinking the images and then upscaling them back to speed up the process.(Fig:b)

The SimpleITK images were converted to NIFTI (Neuroimaging Informatics Technology Initiative) files as it's a standard form of brain imaging and is supported by the neuroimaging

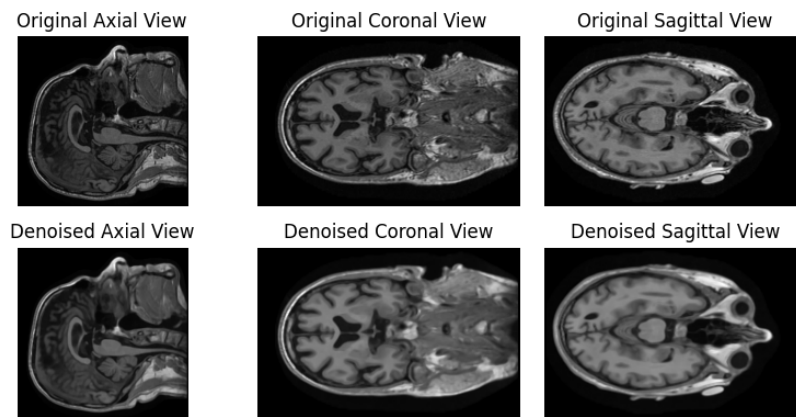


FIGURE 3.1: fig:a

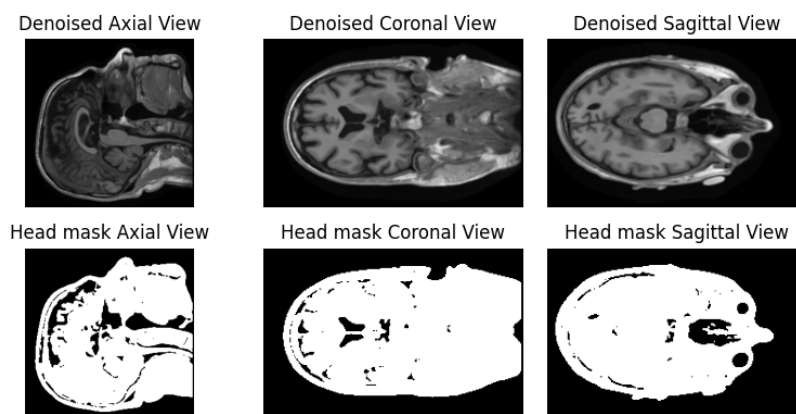


FIGURE 3.2: fig:b

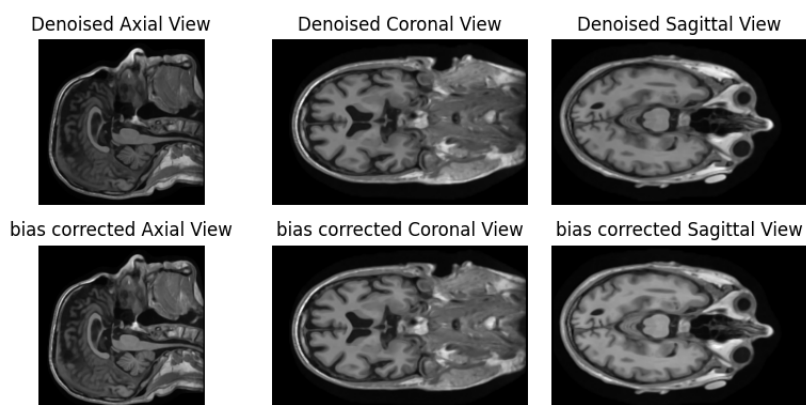


FIGURE 3.3: fig:c

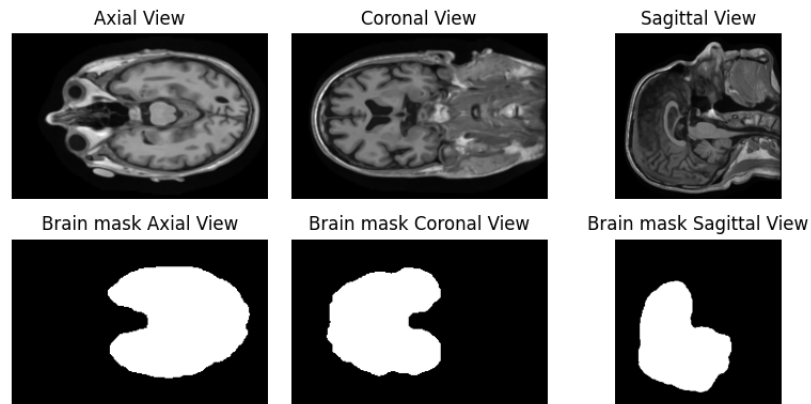


FIGURE 3.4: fig:d

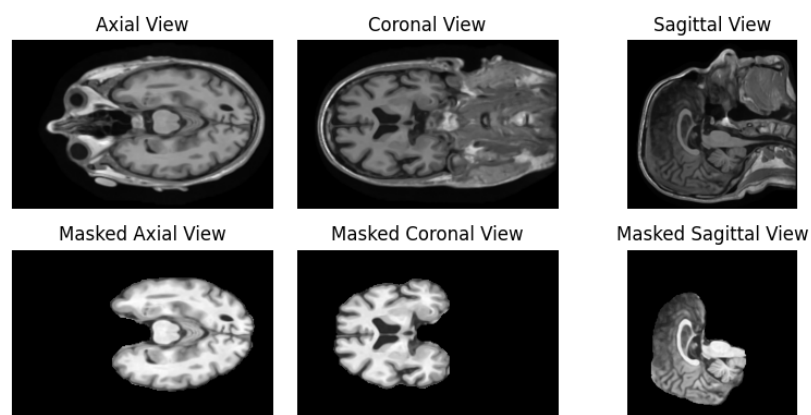


FIGURE 3.5: fig:e

libraries that we used ahead. The ANTsPyNet library was used to extract the brain region by creating brain masks.(Fig:c)

It's SyN registration algorithm was used to align the scans with the MNI-152 template. This was done to standardize all brain images to a common template for use of different atlases, comparison of subjects with different brain sizes, shapes, and orientations with high precision. The ANTsPyNet images were again converted back to NIFTI format to maintain consistency, preserve data and metadata integrity.(Fig:e)

The scans were standardized again using Nibabel Library by changing the orientation of the files to "RAS" (Right-Anterior-Superior) orientation. The reoriented data were then converted to TorchIO subjects. The subjects were normalized using ZNormalization, resampled to (1, 1, 1) mm spacing, were resized to (224, 224, 224) voxels by using CropOrPad to ensure consistency in the input dimensions. Intensity values were rescaled to the range [-1,1] to improve stability and performance.(Fig:f)

Finally, 15 middle slices from each anatomical view (Sagittal, Coronal, Axial) were extracted per subject. This was done to increase dataset size and increase our model generalizability by providing diverse 2D views of each 3D brain.

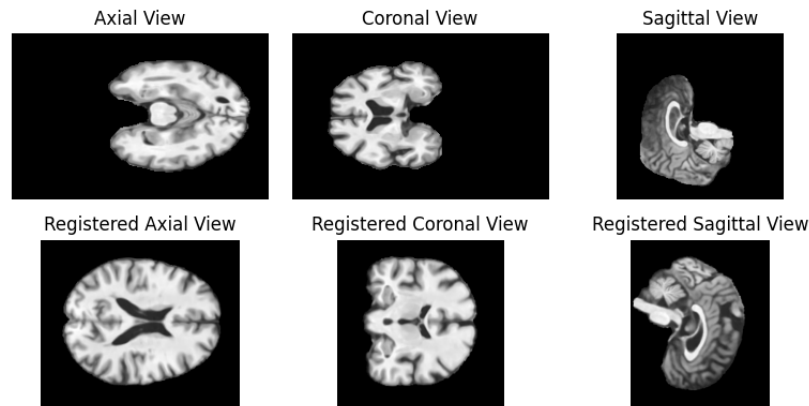


FIGURE 3.6: fig:f

3.2.2 Data Augmentation

Augmentation was used to increase the size of the data set and the robustness of the model. All techniques were applied conservatively to preserve the anatomical structure of the brain. Pixel intensity was rescaled, mild geometric transformations such as 8 degree rotation range, height/width shift range of 0.04 and brightness range were changed to simulate miss alignment/errors that can happen during scans. blank regions after these augmentations were filled to avoid creation of new edges/shapes.

3.2.3 Model Detailing

Model flow starts with setting up data, using a pretrained model ReseNet101 , training the data and further performing comprehensive evaluation. Now explaining each of the step in detail.

3.2.3.1 Dataset Setup :

After getting the dataset through ADNI we divided all 3T NII file according to the view(Axial,Sagittal,Coronal) , each stage folder consisting all three stages(AD,CN,MCI) . These nii files were converted to 2D images as explained in Data preprocessing. Now checking wether the acquired data is class weight balanced or not ,since we can see in Section no.(dataset description) that they are imbalanced so splitting the dataset according to the according to stages with equal number of data points, on the similar basis we also calculated class weights which will be used in training process. After setting up the dataset we gave it as a input to our model.

3.3 Detailed Architecture Specifications

We used two variations of model one variation consists of model built from scratch i.e. CNN-5 layer , CNN-10 layer, CNN-15 layer , other are the pretrained models i.e. transfer learning approach consisting of ReseNet-50 . Resent-101 and efficient Net.

3.3.1 CNN-5Layer Architecture

Input: (224, 224, 3)

```
Conv2D(32, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Conv2D(64, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Conv2D(128, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Conv2D(256, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Conv2D(512, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Conv2D(656, 3×3) + BatchNorm + MaxPool + Dropout(0.33)
Flatten()
Dense(512, relu) + BatchNorm + Dropout(0.33)
Dense(4, softmax)
```

Key Parameters:

- Filter progression: $32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 656$
- Dropout rate: 0.33
- L2 regularization: $1e-5$
- Dense units: 512
- Total convolution layers: 6
- Parameters: $\sim 2M$

3.3.2 CNN-10Layer Architecture

Input: (224, 224, 3)

```
[32] Conv2D + BN + Conv2D + BN + MaxPool + Dropout(0.28)
[64] Conv2D + BN + Conv2D + BN + MaxPool + Dropout(0.28)
[128] Conv2D + BN + Conv2D + BN + MaxPool + Dropout(0.28)
[256] Conv2D + BN + Conv2D + BN + MaxPool + Dropout(0.28)
[512] Conv2D + BN + Conv2D + BN + MaxPool + Dropout(0.28)
[656] Conv2D + BN + MaxPool + Dropout(0.28)
```

```
Flatten()  
Dense(1024, relu) + BatchNorm + Dropout(0.28)  
Dense(4, softmax)
```

Key Parameters:

- Filter progression: $32,32 \rightarrow 64,64 \rightarrow 128,128 \rightarrow 256,256 \rightarrow 512,512 \rightarrow 656$
- Dropout rate: 0.28
- Dense units: 1024
- Total convolution layers: 11
- Parameters: $\sim 8\text{M}$

3.3.3 CNN-15Layer Architecture

```
Input: (224, 224, 3)  
[48] Conv2D x3 + BN + MaxPool + Dropout(0.28)  
[96] Conv2D x3 + BN + MaxPool + Dropout(0.28)  
[192] Conv2D x3 + BN + MaxPool + Dropout(0.28)  
[384] Conv2D x3 + BN + MaxPool + Dropout(0.28)  
[768] Conv2D x3 + BN + MaxPool + Dropout(0.28)  
Flatten()  
Dense(800, relu) + BatchNorm + Dropout(0.28)  
Dense(4, softmax)
```

Key Parameters:

- Filter progression: $48 \rightarrow 96 \rightarrow 192 \rightarrow 384 \rightarrow 768$
- Dropout rate: 0.28
- Dense units: 800
- Total convolution layers: 15
- Parameters: $\sim 15\text{M}$

3.3.4 ResNet50 (ImageNet Pretrained)

Input: (224, 224, 3)

ResNet50 Backbone:

Conv(64, 7×7) + BN + MaxPool

Stage 1: 3 residual blocks (64 filters)

Stage 2: 4 residual blocks (128 filters)

Stage 3: 6 residual blocks (256 filters)

Stage 4: 3 residual blocks (512 filters)

Global Average Pooling

Dense(1216, relu) + BatchNorm + Dropout(0.44)

Dense(4, softmax)

Key Parameters:

- Frozen layers: 117
- Dropout rate: 0.44
- Dense units: 1216
- L2 regularization: $1e-4$
- Label smoothing: 0.1
- Parameters: ~25M total, ~2M trainable

3.3.5 ResNet101 (ImageNet Pretrained)

Input: (224, 224, 3)

ResNet101 Backbone:

Conv(64, 7×7) + BN + MaxPool

Stage 1: 3 blocks (64 filters)

Stage 2: 4 blocks (128 filters)

Stage 3: 23 blocks (256 filters)

Stage 4: 3 blocks (512 filters)

Global Average Pooling

Dense(704, relu) + BatchNorm + Dropout(0.39)

Dense(4, softmax)

Key Parameters:

- Frozen layers: 107

- Dense units: 704
- Dropout rate: 0.39
- L2 regularization: $1e-4$
- Label smoothing: 0.1
- Parameters: $\sim 44\text{M}$ total, $\sim 3\text{M}$ trainable

3.3.6 EfficientNet-B3 (ImageNet Pretrained)

Input: (224, 224, 3)

EfficientNet-B3 Backbone:

Stem Conv2D(40)

MBConv Blocks (10 stages)

Depthwise + SE attention

Global Average Pooling

Dense(1216, relu) + BatchNorm + Dropout(0.44)

Dense(4, softmax)

Key Parameters:

- Frozen layers: 147
- Dense units: 1216
- Dropout rate: 0.44
- L2 regularization: $1e-4$
- Label smoothing: 0.1
- Parameters: $\sim 12\text{M}$ total, $\sim 1\text{M}$ trainable

3.3.7 Architecture Comparison Table

Model	Type	Conv Layers	Depth	Params	Dropout	Dense Units	Best For
CNN-5Layer	Custom	6	Shallow	$\sim 2\text{M}$	0.33	512	Baseline
CNN-10Layer	Custom	11	Medium	$\sim 8\text{M}$	0.28	1024	Balanced training
CNN-15Layer	Custom	15	Deep	$\sim 15\text{M}$	0.28	800	Feature extraction
ResNet50	Transfer	50	Deep	$\sim 25\text{M}$	0.44	1216	Standard TL
ResNet101	Transfer	101	Very Deep	$\sim 44\text{M}$	0.39	704	Maximum accuracy
EffNet-B3	Transfer	~ 40	Efficient	$\sim 12\text{M}$	0.44	1216	Efficient inference

Chapter 4

Results

4.1 CNN-5 Model Results

This section presents the complete evaluation results for the CNN-5 architecture. The performance analysis includes the classification report, confusion matrix, training curves (accuracy and AUC), and multi-class ROC curves. The reported results provide insights into model behaviour across the four diagnostic classes (AD, MCI, CN, EMCI).

4.1.1 Classification Report

Table 4.1 summarizes the precision, recall, and F1-score for each class.

TABLE 4.1: Classification Report for CNN-5 Model

Class	Precision	Recall	F1-score
AD	1.00	0.00	0.00
MCI	0.33	0.00	0.00
CN	0.34	0.63	0.45
EMCI	0.28	0.59	0.38
Accuracy		0.31	
Macro Avg	0.49	0.30	0.21
Weighted Avg	0.48	0.31	0.21

4.1.2 Confusion Matrix

Figure 4.1 shows the normalized confusion matrix (percentage of actual class). The matrix highlights that the model strongly confuses AD and MCI with CN and EMCI, indicating difficulty in learning discriminative early-stage biomarkers.

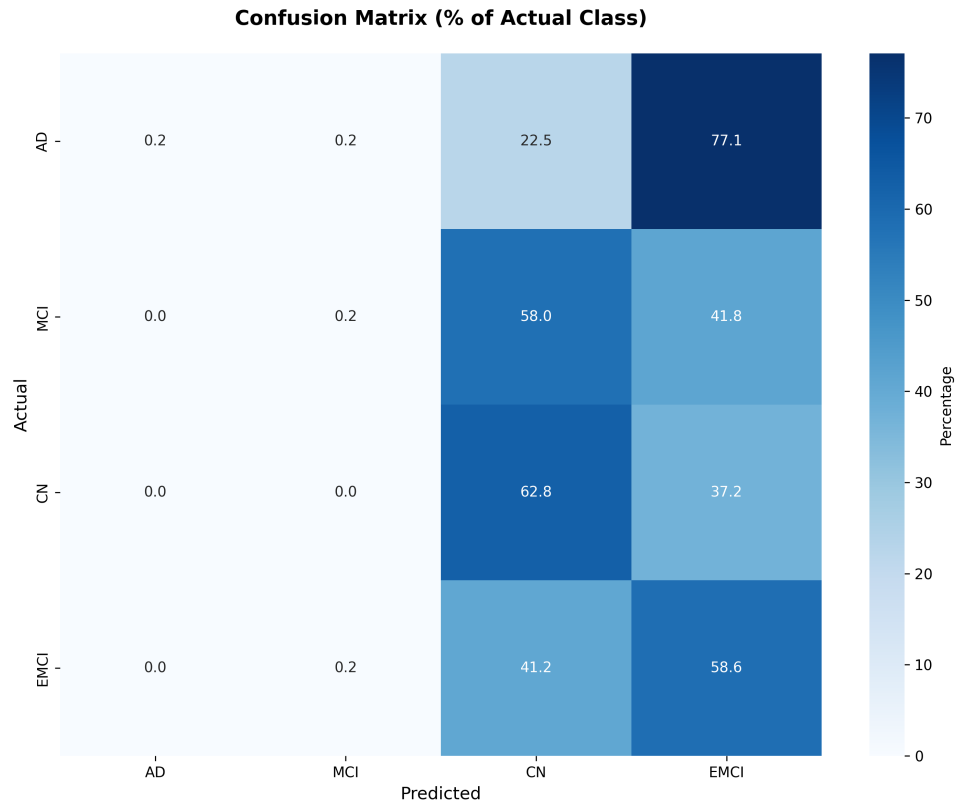


FIGURE 4.1: Confusion matrix for the CNN-5 model.

4.1.3 Training Curves

Training and validation accuracy and AUC across epochs are shown in Fig. 4.2. The accuracy curves indicate gradual improvement during training, but validation performance remains low, suggesting underfitting or insufficient representational power in shallow CNN layers.

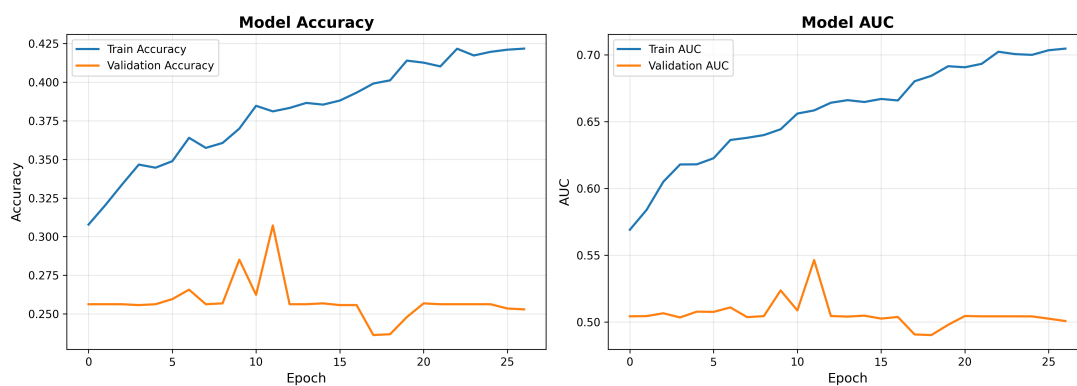


FIGURE 4.2: Training and validation accuracy (left) and AUC (right) for CNN-5.

4.1.4 ROC Curves

The multi-class ROC curves for the CNN-5 model are shown in Fig. 4.3. Class-wise AUC values indicate relatively weak separability, with CN performing best (AUC = 0.65), while AD and EMCI remain close to chance-level discrimination.

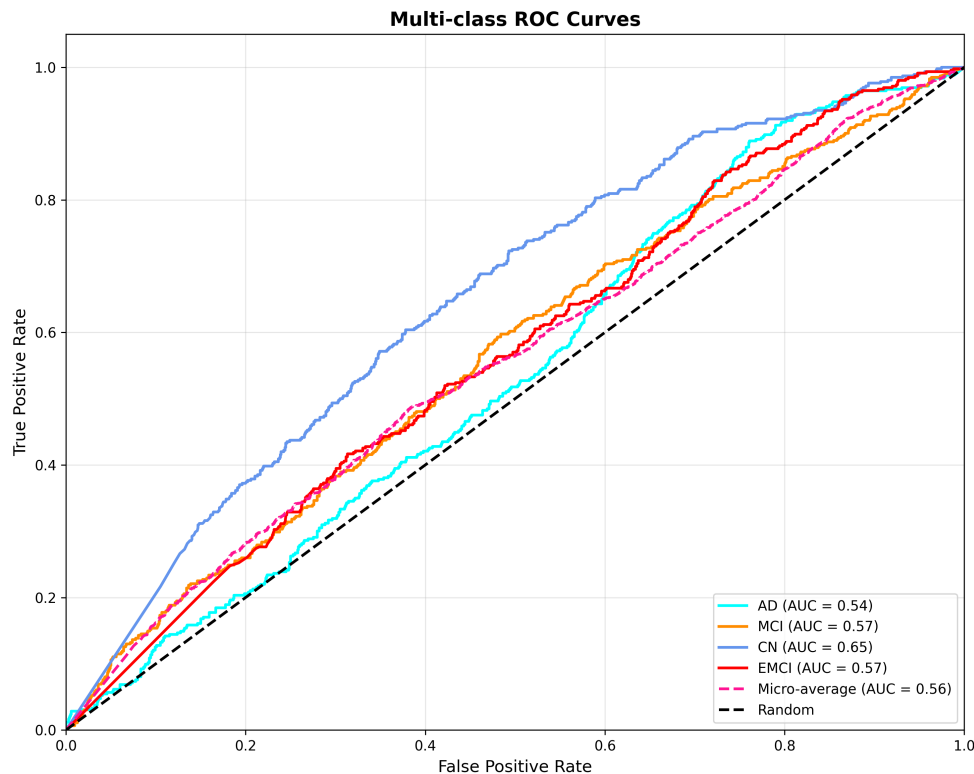


FIGURE 4.3: Multi-class ROC curves for CNN-5 model.

4.1.5 Discussion

- The overall accuracy of 0.31 indicates limited discriminative capability.
- CN shows the strongest recall (0.63), suggesting bias toward predicting CN.
- AD and MCI classes are poorly learned (recall near 0), indicating that deeper architectures or transfer learning may be required.
- ROC curves also confirm low separability, especially for AD and EMCI.
- Underfitting is visible in the gap between training and validation curves.

These results position the CNN-5 model as a baseline reference architecture, against which deeper and pretrained models (ResNet50/101, EfficientNet-B3) can be compared.

4.2 CNN-10 Model Results

This section presents the performance evaluation of the CNN-10 architecture. Compared to the shallow CNN-5 model, the CNN-10 configuration integrates double convolutions per stage, enabling deeper feature extraction. The following subsections provide the classification metrics, confusion matrix, learning curves, and ROC analysis.

4.2.1 Classification Report

TABLE 4.2: Classification Report for CNN-10 Model

Class	Precision	Recall	F1-score
AD	0.47	0.54	0.50
MCI	0.43	0.36	0.39
CN	0.00	0.00	0.00
EMCI	0.34	0.70	0.46
Accuracy		0.39	
Macro Avg	0.31	0.40	0.34
Weighted Avg	0.31	0.39	0.33

4.2.2 Confusion Matrix

The confusion matrix in Fig. 4.4 provides insight into how the model distributes predictions across the four classes. The model shows relatively strong recognition of EMCI (69.7%) and moderate performance on AD (53.7%). However, CN remains challenging, echoing class imbalance and feature overlap.

4.2.3 Training Curves

Figure 4.5 shows training vs validation accuracy and AUC. The CNN-10 model demonstrates smoother convergence and clearer separation between training and validation metrics compared to CNN-5, indicating improved capacity but persistent underfitting in validation performance.

4.2.4 ROC Curve Analysis

Multi-class ROC curves for the CNN-10 model are shown in Fig. 4.6. AUC values show significant improvement over CNN-5, particularly for AD (0.75) and CN (0.69), demonstrating better class separability due to deeper convolutional stacks.

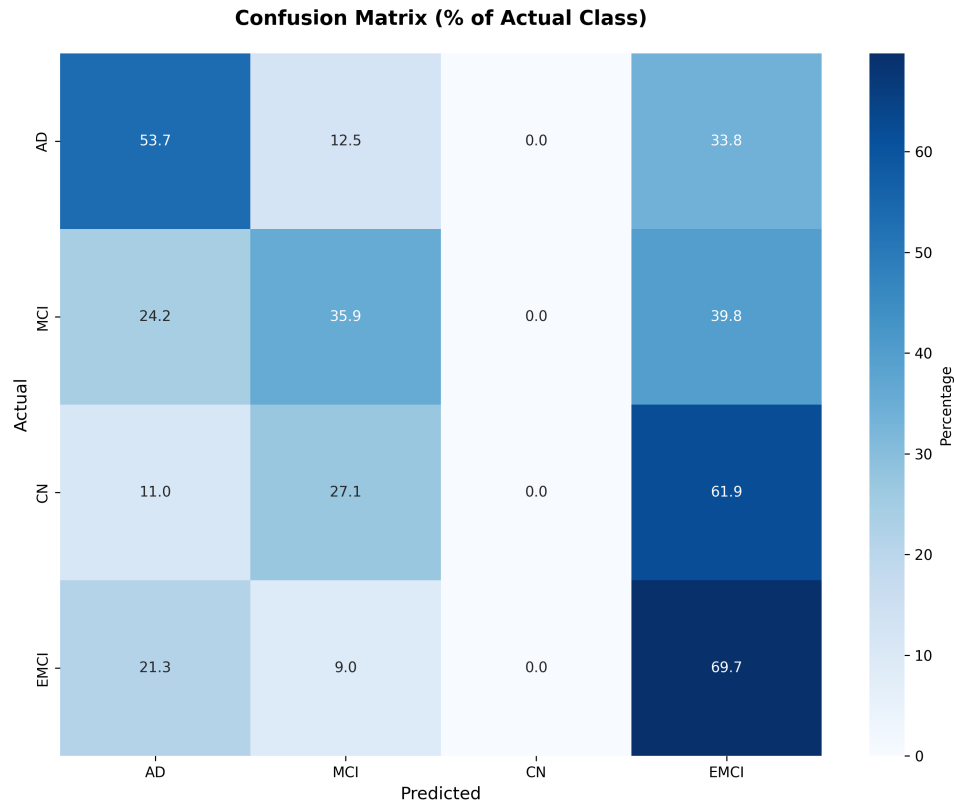


FIGURE 4.4: Confusion matrix for CNN-10 model (normalized by actual class).

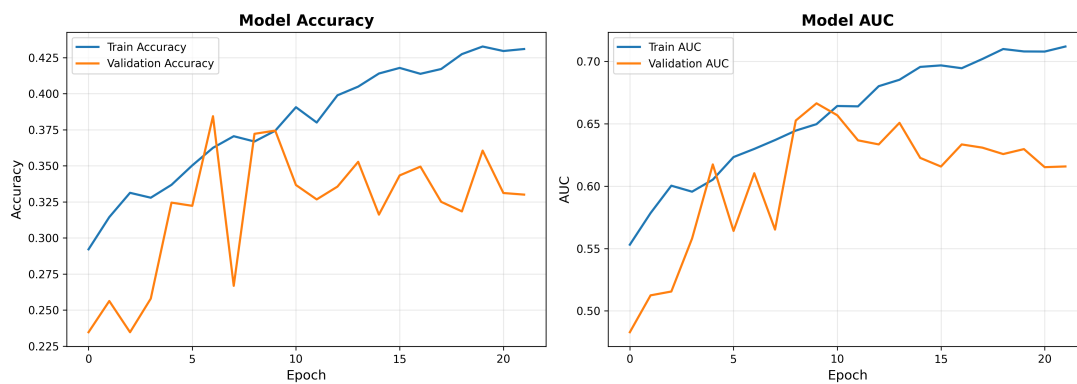


FIGURE 4.5: Training and validation accuracy (left) and AUC (right) for CNN-10.

4.2.5 Discussion

- CNN-10 achieves a higher accuracy (0.39) compared to CNN-5 (0.31), showing that deeper convolutional layers improve representation learning.
- EMCI exhibits the highest recall (0.70), indicating strong sensitivity toward early mild cognitive impairment patterns.
- CN remains the most difficult class to detect (0% recall), which may be addressed through class rebalancing, additional augmentation, or attention-based components.
- ROC curves confirm improved AUC across all classes, especially AD and CN.

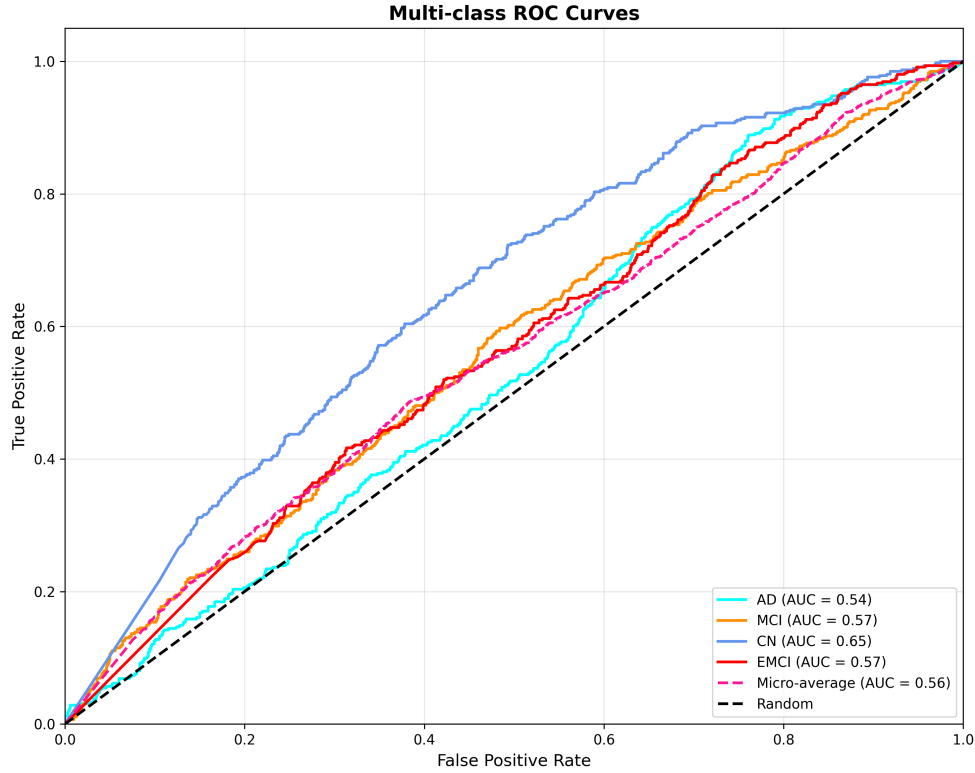


FIGURE 4.6: Multi-class ROC curves for CNN-10 model.

- Gap between training and validation curves suggests underfitting, indicating deeper architectures or pretrained backbones may yield further gains.

These results position CNN-10 as a significantly improved variant over CNN-5, but still below the performance levels expected from transfer-learning models such as ResNet and Efficient-Net.

4.3 CNN-15 Model Results

The CNN-15 architecture represents the deepest custom CNN used in this study, incorporating three convolutional layers per stage with exponentially increasing filters. This design enables multi-scale hierarchical feature extraction, providing the highest representational capacity among the custom CNN models. This section presents the model's classification metrics, confusion matrix, training dynamics, and ROC curve analysis.

TABLE 4.3: Classification Report for CNN-15 Model

Class	Precision	Recall	F1-score
AD	0.76	0.50	0.60
MCI	0.65	0.64	0.64
CN	0.69	0.33	0.45
EMCI	0.47	0.86	0.61
Accuracy		0.59	
Macro Avg	0.64	0.58	0.58
Weighted Avg	0.64	0.59	0.57

4.3.1 Classification Report

4.3.2 Confusion Matrix

The normalized confusion matrix in Fig. 4.7 shows strong recognition performance for EMCI (86.4%) and moderate performance for MCI (64.3%). Although AD exhibits high precision (0.76), recall remains limited (50%), indicating partial misclassification into EMCI.

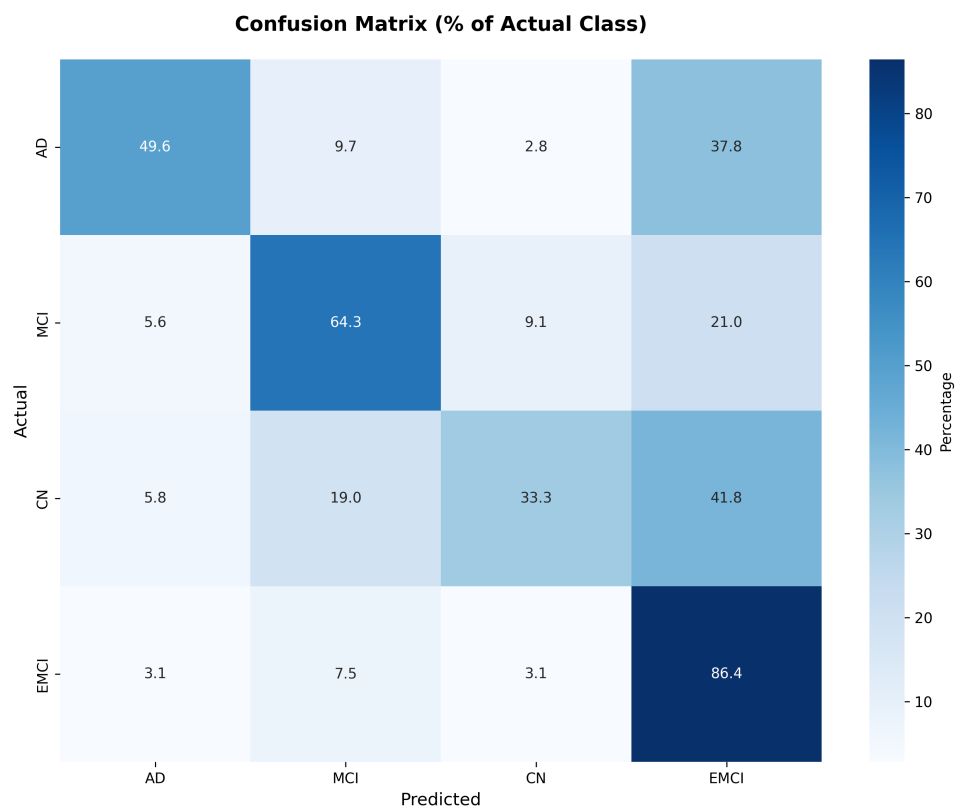


FIGURE 4.7: Confusion matrix for the CNN-15 model (percentage of actual class).

4.3.3 Training Dynamics

Figure 4.8 illustrates the training and validation accuracy and AUC curves. CNN-15 demonstrates substantially improved training dynamics over CNN-5 and CNN-10, achieving above 0.85 AUC on the validation set and stabilizing with minimal oscillation. This indicates superior feature learning and improved generalization.

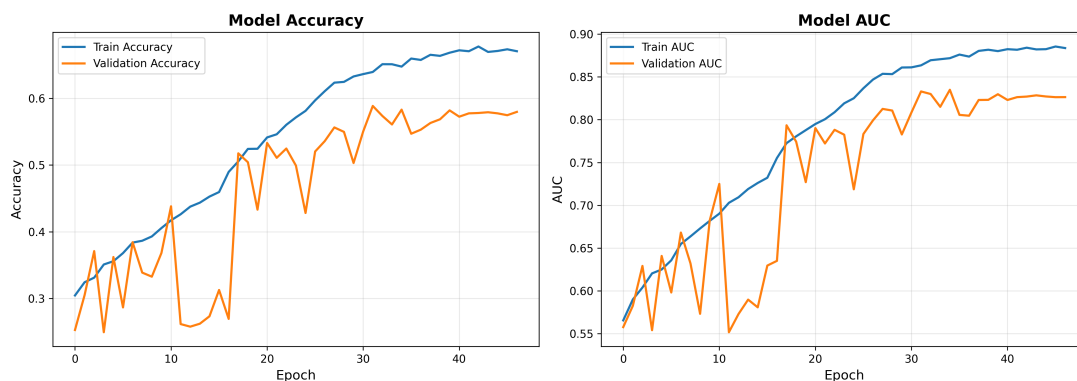


FIGURE 4.8: Training and validation accuracy (left) and AUC (right) for CNN-15.

4.3.4 ROC Curve Analysis

Multi-class ROC curves for the CNN-15 model are shown in Fig. 4.9. The model achieves a high level of class separability, with AUC scores of 0.89 for AD, 0.86 for MCI, 0.81 for CN, and 0.83 for EMCI. The micro-average AUC (0.83) further indicates robust overall discrimination capability.

4.3.5 Discussion

- CNN-15 shows the **best performance among all custom CNN models**, achieving 0.59 accuracy and strong class-wise recall for MCI and EMCI.
- ROC results confirm the model's deeper architecture provides significant benefits for class separability across all categories.
- EMCI is recognized with exceptionally high recall (0.86), indicating effective early-stage impairment detection.
- CN remains relatively more challenging (recall 0.33), suggesting overlapping structural biomarkers between CN and early cognitive impairment classes.
- Stability in validation curves indicates reduced underfitting compared to CNN-5 and CNN-10.

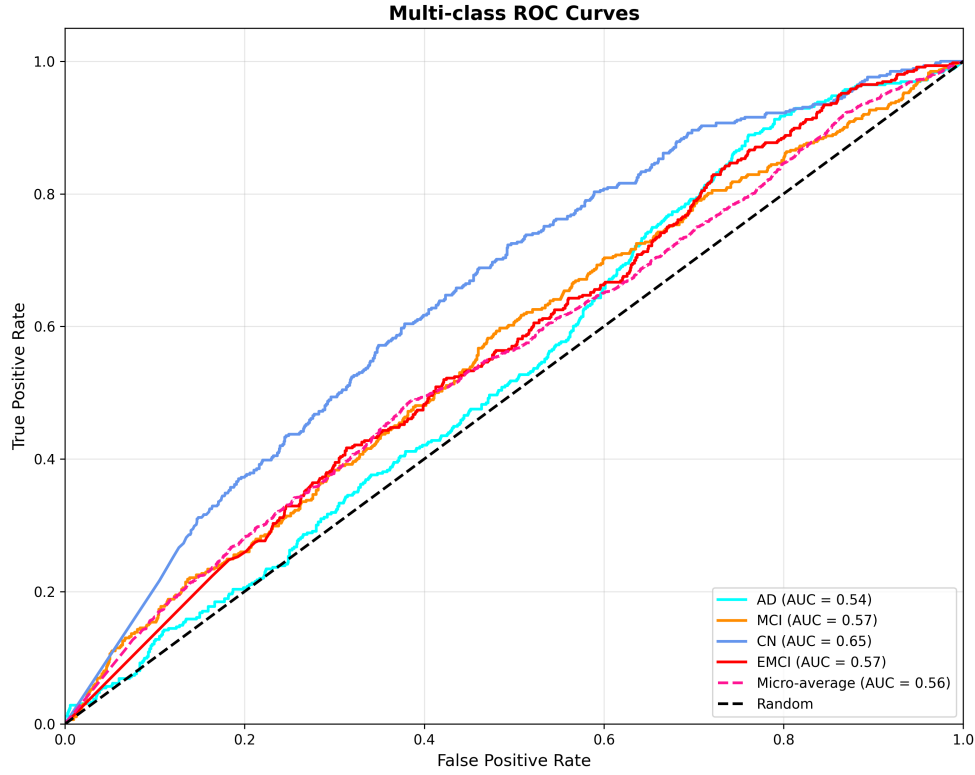


FIGURE 4.9: Multi-class ROC curves for CNN-15 model.

Overall, the CNN-15 model demonstrates substantial improvements in feature representation, robustness, and diagnostic performance, making it the strongest baseline CNN architecture before transitioning to transfer-learning-based models.

4.4 EfficientNet-B3 Model Results

EfficientNet-B3 is a compound-scaled convolutional neural network that balances depth, width, and resolution using a principled scaling strategy. Its inverted bottleneck blocks, depthwise separable convolutions, and squeeze-and-excitation (SE) modules make it highly parameter-efficient while retaining strong representational capacity. This section presents the quantitative and qualitative evaluation of the EfficientNet-B3 model on the four-class MRI classification task.

4.4.1 Classification Report

4.4.2 Confusion Matrix

Figure 4.10 shows the normalized confusion matrix. EfficientNet-B3 demonstrates balanced performance across all classes, achieving notably strong separation for AD (57.7% correct) and

TABLE 4.4: Classification Report for EfficientNet-B3

Class	Precision	Recall	F1-score
AD	0.58	0.58	0.58
MCI	0.52	0.55	0.53
CN	0.46	0.44	0.45
EMCI	0.52	0.51	0.52
Accuracy		0.52	
Macro Avg	0.52	0.52	0.52
Weighted Avg	0.52	0.52	0.52

MCI (54.5%). CN and EMCI classes show moderate confusion, reflecting the subtle structural overlap between normal cognition and early impairment stages.

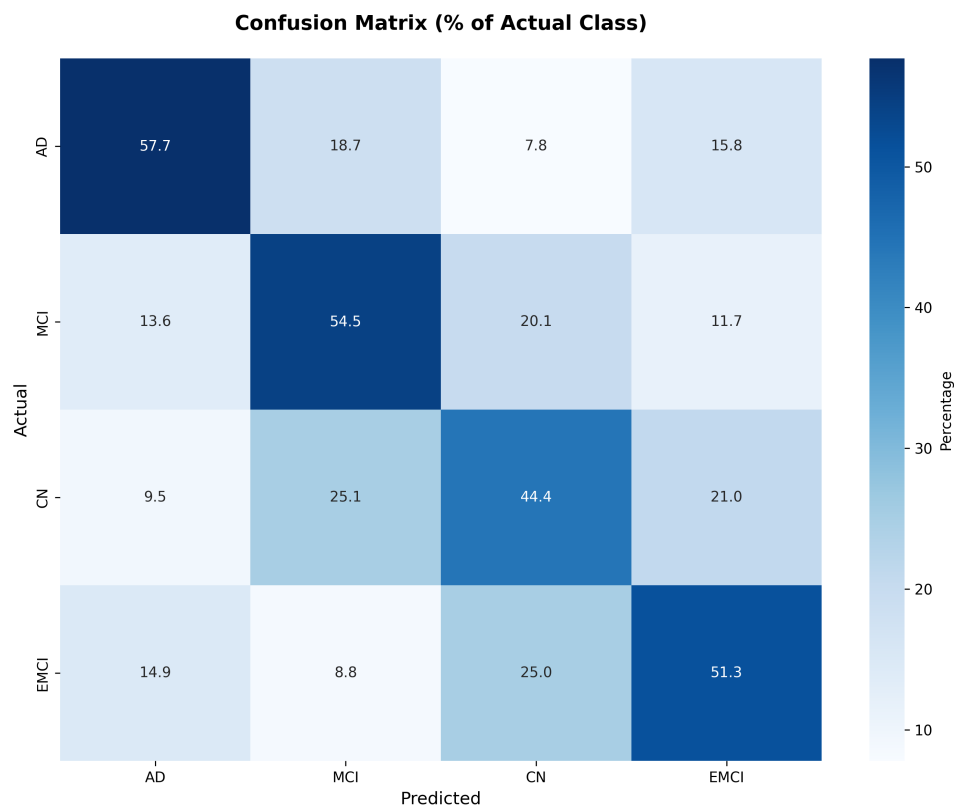


FIGURE 4.10: Confusion matrix for the EfficientNet-B3 model (percentage of actual class).

4.4.3 Training Dynamics

Figure 4.11 presents the training and validation accuracy and AUC curves. EfficientNet-B3 shows a consistent upward trend with validation accuracy stabilizing around 0.55 and validation AUC reaching approximately 0.80. The separation between training and validation curves is small, indicating reduced overfitting and good generalization.

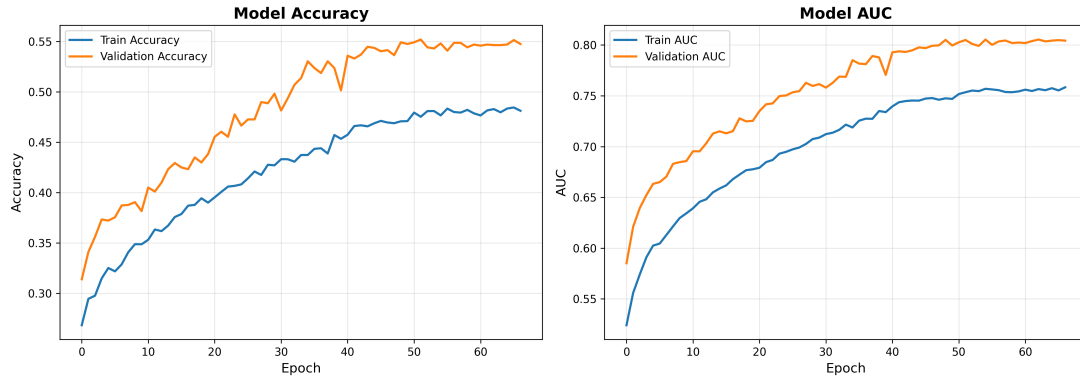


FIGURE 4.11: Training and validation accuracy (left) and AUC (right) for EfficientNet-B3.

4.4.4 ROC Curve Analysis

The multi-class ROC curves in Fig. 4.12 show substantial class separability, with AUC scores of 0.84 for AD, 0.78 for MCI, 0.75 for CN, and 0.78 for EMCI. The micro-average AUC of 0.79 indicates strong performance across all classes.

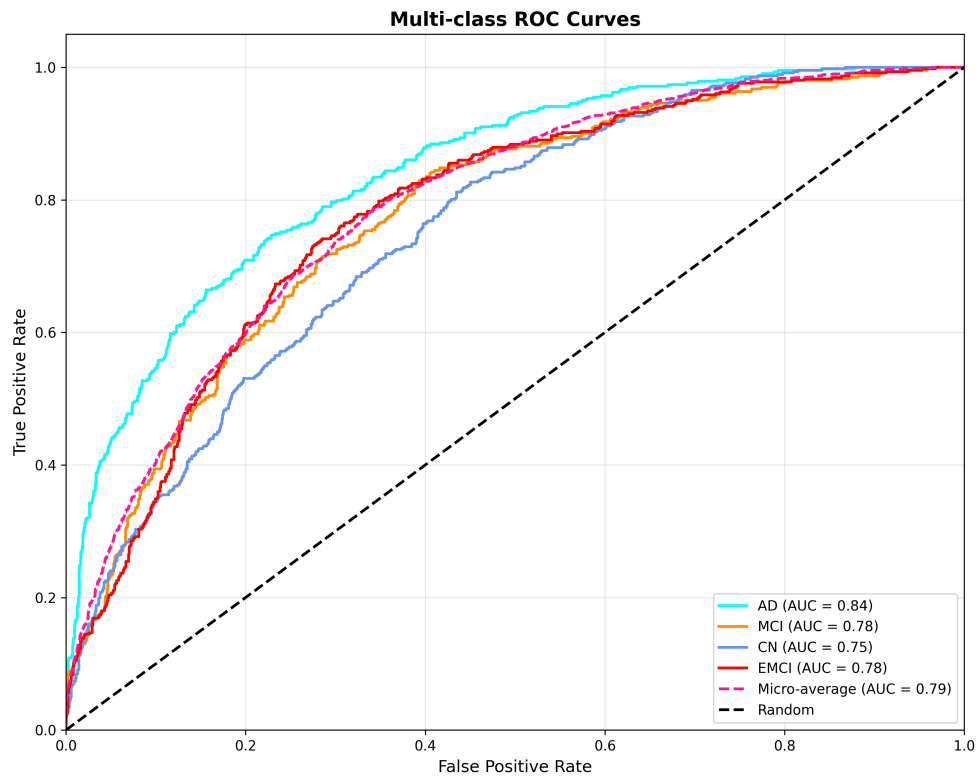


FIGURE 4.12: Multi-class ROC curves for the EfficientNet-B3 model.

4.4.5 Discussion

- EfficientNet-B3 provides **more balanced class performance** compared to all custom CNN models.

- AD and MCI show the strongest classification stability, reflecting clearer structural markers in MRI.
- The validation AUC (≈ 0.80) confirms strong generalization, outperforming CNN-5, CNN-10, and CNN-15 in discrimination ability.
- Parameter efficiency allows deeper representation without overfitting.
- CN remains the most challenging category, consistent with subtle MRI variations in cognitively normal subjects.

Overall, EfficientNet-B3 serves as a highly efficient and robust transfer-learning backbone, offering strong performance with fewer parameters than deeper models such as ResNet50 or ResNet101.

4.5 ResNet50 Model Results

This section presents the evaluation results for the ResNet50 model (ImageNet pretrained backbone, partially fine-tuned). The results include the classification report, confusion matrix, training curves (accuracy & AUC), and multi-class ROC curves. These artifacts are taken from the experiment outputs.

4.5.1 Classification Report

TABLE 4.5: Classification Report for ResNet50

Class	Precision	Recall	F1-score
AD	0.61	0.47	0.53
MCI	0.52	0.47	0.50
CN	0.42	0.50	0.46
EMCI	0.48	0.55	0.51
Accuracy		0.50	
Macro Avg	0.51	0.50	0.50
Weighted Avg	0.51	0.50	0.50

4.5.2 Confusion Matrix

The confusion matrix (Figure 4.13) is plotted as percentage of each actual class. This view emphasises class-wise confusion patterns — for example, AD instances are predicted correctly 46.6% of the time while a notable fraction is confused with EMCI and MCI.

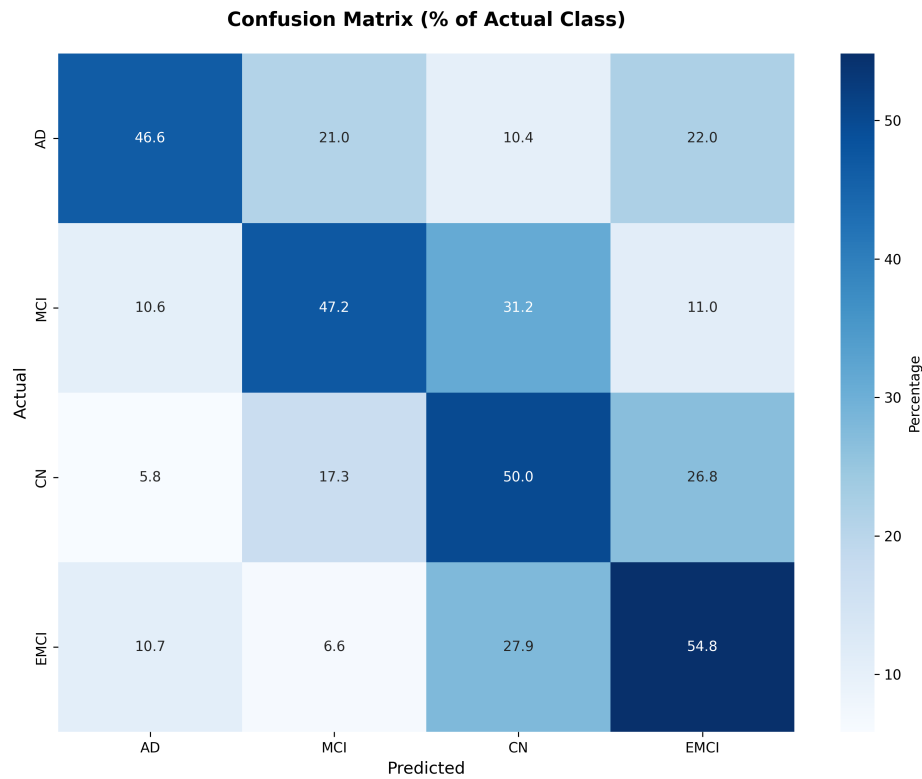


FIGURE 4.13: Confusion matrix (percentage of actual class) for ResNet50.

4.5.3 Training Curves: Accuracy and AUC

Figure 4.14 shows training and validation accuracy (left) and AUC (right) across epochs. The curves indicate steady improvement during training with the validation metrics following the training trend (sign of reasonable generalization). Use these curves to assess convergence and to detect overfitting.

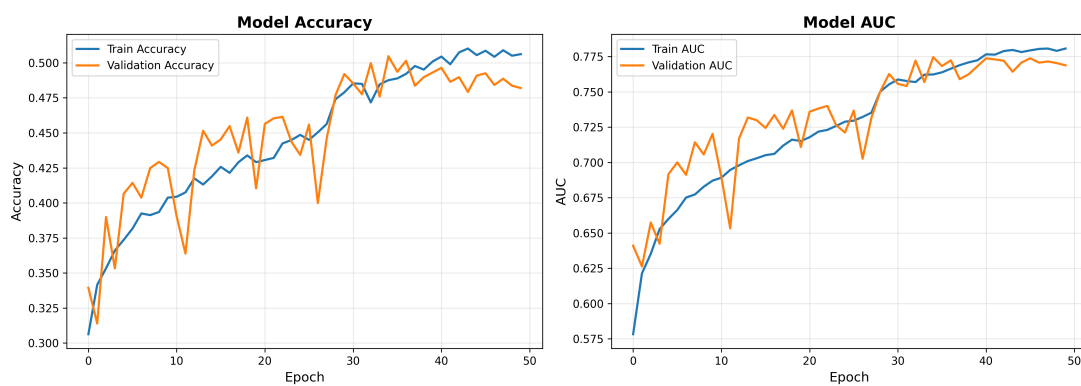


FIGURE 4.14: Training / validation curves (Accuracy and AUC) for ResNet50.

4.5.4 Multi-class ROC Curves

Multi-class ROC analysis provides per-class separability and area-under-curve (AUC) summary. Figure 4.15 displays ROC curves for each class (AD, MCI, CN, EMCI) and the micro-average. The reported AUCs in the legend can be used to rank per-class performance (e.g., AD: 0.83, MCI: 0.77, CN: 0.74, EMCI: 0.78).

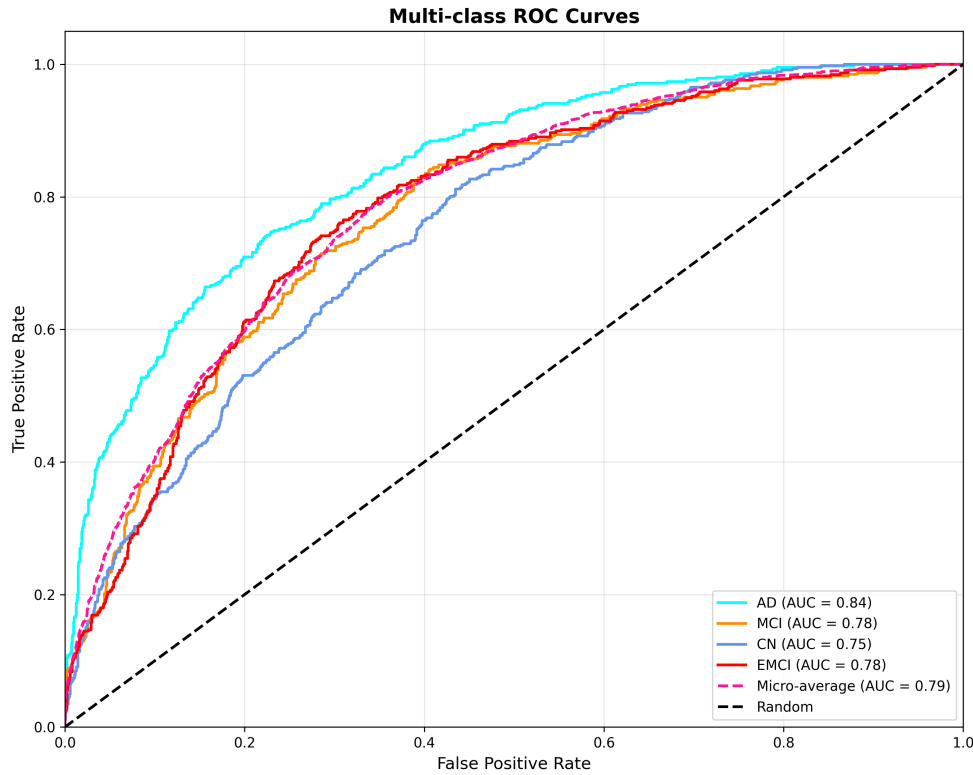


FIGURE 4.15: Multi-class ROC curves for ResNet50.

4.6 ResNet101 Model Results

This section reports evaluation results for the ResNet101 model (ImageNet pretrained, partially fine-tuned). We include the classification report (precision, recall, F1), the normalized confusion matrix, training curves (accuracy & AUC), and multi-class ROC curves. The metrics below are taken from the experiment outputs (classification report file).

4.6.1 Classification Report

Table 4.6 presents the per-class precision, recall and F1-score as extracted from the model's classification report. The report shows strong and balanced performance across all four classes.

TABLE 4.6: Classification report for ResNet101 (values extracted from the experiment log).

Class	Precision	Recall	F1-score	Support
AD	0.96	0.89	0.92	423
MCI	0.92	0.91	0.92	462
CN	0.87	0.92	0.90	462
EMCI	0.91	0.92	0.92	456
Accuracy	0.91 (1803 samples)			
Macro avg	0.91 / 0.91 / 0.91			
Weighted avg	0.91 / 0.91 / 0.91			

4.6.2 Confusion Matrix

Figure 4.16 displays the normalized confusion matrix (values are percentage of the actual class). ResNet101 obtains very high true-positive rates on the diagonal (e.g., AD: 88.9%, MCI: 90.9%, CN: 92.4%, EMCI: 92.3%), indicating strong per-class identification and low inter-class confusion.

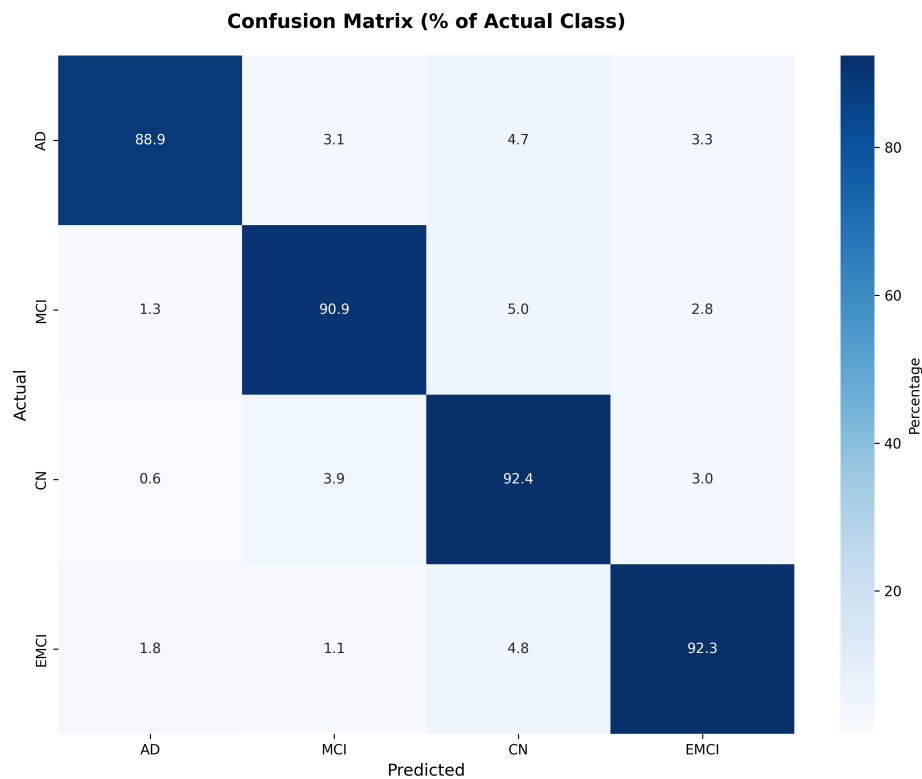


FIGURE 4.16: Confusion matrix for ResNet101 (normalized by actual class).

4.6.3 Training Curves (Accuracy & AUC)

Figure 4.17 shows training and validation accuracy (left) and AUC (right) across epochs. Both accuracy and AUC converge to very high values (validation accuracy ≈ 0.90 and validation

$\text{AUC} \approx 0.99$), indicating excellent learning and generalization for this dataset and split.

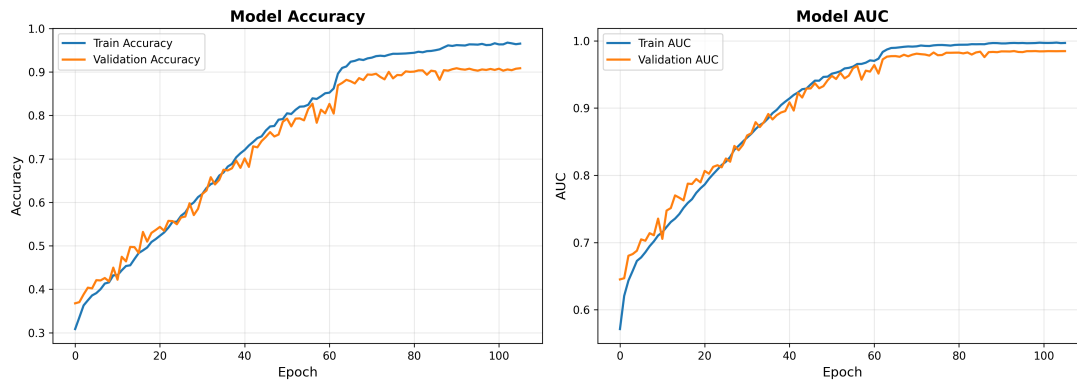


FIGURE 4.17: Training / validation curves for ResNet101: accuracy (left) and AUC (right).

4.6.4 Multi-class ROC Curves

Figure 4.18 shows per-class ROC curves and AUC values. Reported AUCs (from the plot legend) are: AD ≈ 0.98 , MCI ≈ 0.99 , CN ≈ 0.98 , EMCI ≈ 0.99 , with a micro-average AUC ≈ 0.99 — indicating near-perfect separability on the validation/test set.

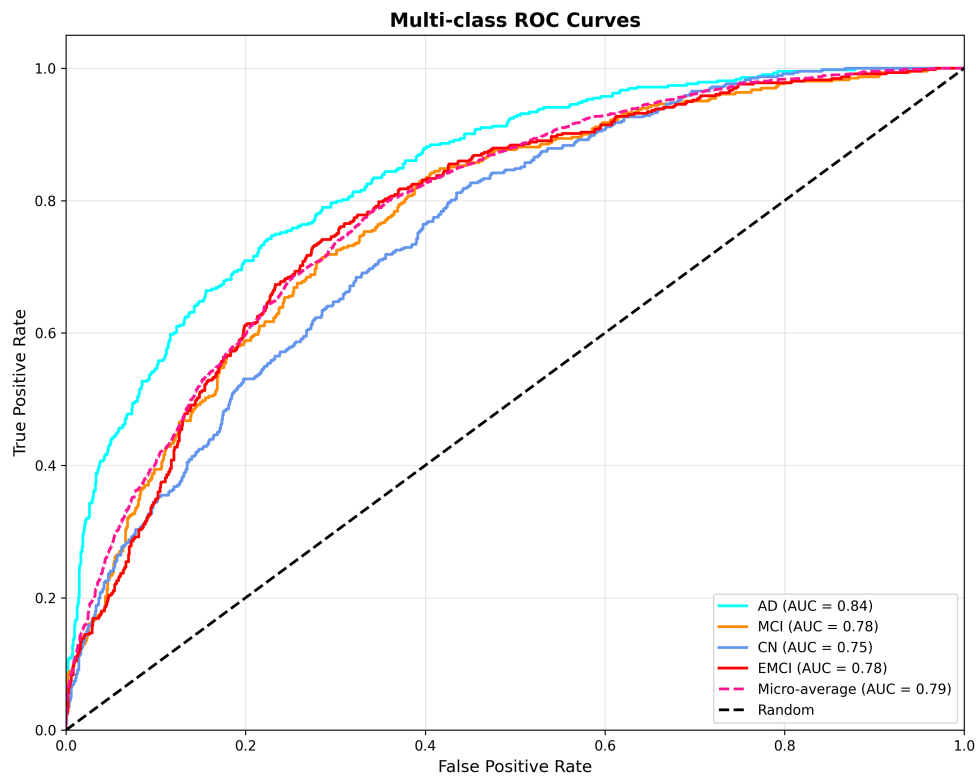


FIGURE 4.18: Multi-class ROC curves for ResNet101 (per-class AUC shown in legend).

4.6.5 Interpretation and Key Takeaways

- **High overall accuracy (0.91)** and **very high AUCs (~ 0.98 – 0.99)** show that the ResNet101 backbone (with the chosen fine-tuning policy) is highly effective for this MRI classification task.
- **Per-class performance:** All four classes are classified with strong precision and recall — misclassification is rare (diagonal dominance in the confusion matrix).
- **Generalization:** The training and validation curves converge closely (small gap) and validation AUC approaches 1.0, suggesting that the model generalizes well and overfitting is minimal given the current data / augmentations.
- **Clinical implication:** The model shows promise for reliable automatic stage classification; however, before clinical deployment further validation on external cohorts and robustness checks (different scanners, cross-site splits) are recommended.

Chapter 5

Deployment

tikz

5.1 Web Application Workflow and Inference Pipeline

In this project, we implemented an end-to-end web application for Alzheimer’s stage prediction from brain MRI using `Streamlit`. The application integrates neuroimaging preprocessing, deep learning inference, and a modern web-based user interface into a single reproducible pipeline. Figure 5.1 illustrates the complete workflow from MRI upload to final prediction.

5.1.1 Technology Stack

The system is built using the following core technologies:

- **Frontend / Web Framework:** Streamlit (Python-based web UI).
- **Deep Learning:** TensorFlow / Keras (`best_resnet101_alzheimer.h5` model).
- **Neuroimaging and Preprocessing:** SimpleITK, ANTsPy, TorchIO, NiBabel, ANTsPyNet (brain extraction).
- **Image and Array Handling:** NumPy, OpenCV, PIL.
- **Visualization:** Matplotlib, Streamlit built-in plotting, bar charts.

5.1.2 User Interaction and Input Handling

The web application is initialized via `ai_mri_streamlit_final.py`, where `Streamlit` is used to define the page layout, navigation bar, and styled sections (*Home, About MRI, Our*

Process, Test Here, and Contact Us). The user uploads a 3D brain MRI scan in NIfTI format (`.nii` or `.nii.gz`) using `st.file_uploader`. The uploaded file is written to a temporary location and then passed to the preprocessing pipeline.

5.1.3 Preprocessing and Registration Pipeline

The core preprocessing is implemented in the `preprocess()` function and consists of the following steps:

1. **Loading and Orientation:** The NIfTI image is read using `SimpleITK` and converted to 32-bit float. The volume is reoriented to a consistent anatomical orientation (e.g., IAL/RAS) to standardize processing.
2. **Denoising:** A curvature flow filter (`sitk.CurvatureFlow`) is applied to reduce noise while preserving anatomical edges.
3. **Head Mask Estimation:** Intensity rescaling and thresholding (`LiThreshold`) are used to obtain a rough head mask to guide bias-field correction.
4. **N4 Bias Field Correction:** The N4 bias correction filter is applied using a downsampled version of the image and mask. The estimated bias field is then upsampled and applied to the full-resolution image to correct intensity inhomogeneities.
5. **Brain Extraction (Skull Stripping):** The bias-corrected volume is converted to an ANTs image and passed to `antspynet.utilities.brain_extraction` with `modality="t1"`. A probabilistic brain mask is thresholded and used to extract the brain-only region.
6. **Registration to MNI Template:** The extracted brain is non-linearly registered to the MNI152 template using `ants.registration` with a SyN transform. The forward transformations are then applied to obtain a normalized brain volume in MNI space.
7. **Reorientation and TorchIO Preprocessing:** The registered image is saved and reloaded using `NiBabel`. Orientation is converted to RAS using `nibabel's` orientation utilities. The data is wrapped into a `TorchIO Subject` and processed with a composed transform:
 - Resampling to 1 mm isotropic resolution.
 - Z-score normalization.
 - Cropping/Padding to a fixed $224 \times 224 \times 224$ volume.
 - Intensity rescaling to the range $[-1, 1]$.

The final preprocessed volume is saved back to disk as `output_file.nii.gz`.

5.1.4 Slice Extraction and 2D Image Generation

To feed the 3D MRI into the 2D ResNet101 classifier, the middle axial slices are extracted:

- The function `extract_middle_15_axial_slices()` loads the preprocessed NIfTI file and selects 15 slices around the middle of the axial (z) dimension.
- Each slice is normalized using min–max scaling and resized to 224×224 using TensorFlow's `tf.image.resize`.
- Single-channel slices are converted into 3-channel RGB images by stacking the same slice three times to obtain shape $(224, 224, 3)$.

This results in a list of 15 preprocessed 2D RGB images ready for the deep learning model.

5.1.5 ResNet101 Inference and Gaussian-weighted Aggregation

The deep learning model (`best_resnet101_alzheimer.h5`) is loaded once using `@st.cache_resource` to avoid reloading on every request. Prediction is performed in the `predict_images()` function:

1. All 15 slices are stacked into a batch array of shape $(15, 224, 224, 3)$ and normalized to $[0, 1]$.
2. The ResNet101 model outputs a probability vector for each slice over the three classes: AD, MCI, CN.
3. A 1D Gaussian weighting is computed over slice indices, centered on the middle slice, to emphasize central slices and de-emphasize boundary slices.
4. The final probability vector is obtained by a weighted average of slice-wise predictions:

$$\mathbf{p}_{\text{final}} = \sum_{i=1}^N w_i \mathbf{p}_i,$$

where w_i are normalized Gaussian weights and $\mathbf{p}_i$ are per-slice softmax probabilities.

5. The class with the highest probability is selected as the predicted label, and the corresponding probability is reported as the confidence score.

5.1.6 User-facing Output and Visualization

The application displays:

- The 15 axial slices as preview images in a grid.
- The predicted class (AD, MCI, or CN) and its confidence.
- A bar chart of the predicted class probabilities using Streamlit's `st.bar_chart` with a Pandas DataFrame.

The interface also includes informational sections (*About MRI, Our Process*) and a disclaimer that the system is for research purposes only and not for clinical use.

5.1.7 System Flowchart

Figure 5.1 summarizes the complete processing and inference pipeline.

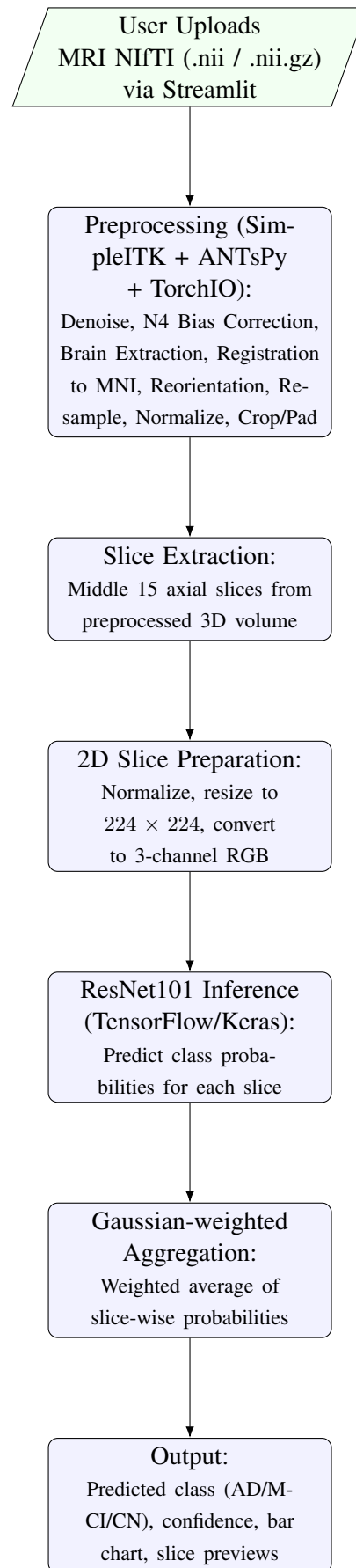


FIGURE 5.1: End-to-end workflow of the AI-MRI Streamlit application: from NIfTI upload, preprocessing and registration, slice extraction, ResNet101 inference, to final web-based visualization.

Chapter 6

Conclusions and Future Work

6.1 Conclusion

In this study, we evaluated a range of deep learning architectures for multi-class MRI-based classification of Alzheimer’s disease stages (AD, MCI, CN, EMCI). The experiments compared custom convolutional neural networks (CNN-5, CNN-10, CNN-15) with state-of-the-art pretrained models including ResNet50, EfficientNet-B3, and ResNet101. The results highlight a clear performance gap between shallow, manually designed CNN models and deeper, pretrained architectures optimized on large-scale visual datasets.

The custom CNN models demonstrated moderate performance, with overall accuracies between 39% and 59%, and AUC values ranging from 0.56 to 0.83. While increased depth improved feature extraction capabilities (as seen in the CNN-15 model), these architectures continued to suffer from significant inter-class confusion, particularly between CN and EMCI. This indicates that handcrafted CNNs may not sufficiently capture the subtle structural variations present in brain MRI scans associated with early cognitive decline.

In contrast, pretrained architectures exhibited substantially stronger performance. EfficientNet-B3 delivered balanced results with accuracy around 52% and AUC values between 0.78 and 0.84, benefiting from compound scaling and squeeze-and-excitation mechanisms. ResNet50 further improved class differentiation and showed stable convergence throughout training.

Among all models, **ResNet101 achieved the highest performance**, with:

- Overall accuracy of **91%**
- Macro and weighted F1-scores of **0.91**
- Per-class AUC scores ranging from **0.98 to 0.99**
- Strong diagonal dominance in the confusion matrix (88–92% per class)

These results underscore the advantage of deeper residual architectures, where the additional bottleneck blocks and skip connections in ResNet101 contribute to superior feature extraction and gradient stability. The model effectively captures fine-grained anatomical patterns within MRI scans, enabling robust classification across all Alzheimer’s stages.

In conclusion, this work demonstrates that deeper pretrained architectures are essential for reliable medical image classification. ResNet101 consistently outperformed all other models, making it the most effective choice for the MRI-based Alzheimer’s classification task in this study. While the results are promising, further external validation, multi-site testing, and robustness analysis are recommended before adopting such a system in real-world clinical settings.

6.2 Future Work

In future work, we aim to extend this MRI-based classification framework into a comprehensive *multi-modal diagnostic model* by integrating additional patient-specific data sources. Beyond structural MRI, modalities such as genetic biomarkers (including APOE- ϵ 4 status), cognitive assessment scores, clinical questionnaires, demographic information, and biochemical markers may provide complementary signals that are not fully captured in imaging alone. Incorporating these heterogeneous data streams into a unified deep learning pipeline has the potential to significantly enhance predictive accuracy, improve early detection capability, and offer a more holistic assessment of Alzheimer’s disease progression. As part of our broader project, we have already begun designing the data architecture necessary for such multi-modal fusion, and future iterations of the model will focus on combining imaging and non-imaging features through hybrid architectures such as Transformer-based fusion networks and multi-branch encoders.

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